Exercices I.2

#1 a) $\langle x, x\rangle = \sum x_i^2 \geq 0$

$\langle x, x\rangle = 0 \iff \sum x_i^2 = 0 \iff x_i^2 = 0 \ (\forall i) \iff x = 0$

b) $\langle x, y\rangle = \sum x_i y_i = \sum y_i x_i = \langle y, x\rangle$

c) $\langle x, \alpha_1 y_1 + \alpha_2 y_2\rangle = \sum x_i (\alpha_1 y_1 + \alpha_2 y_2)_i = \sum x_i (\alpha_1 y_{1_i} + \alpha_2 y_{2_i})$

$= \alpha_1 \sum x_i y_{1_i} + \alpha_2 \sum x_i y_{2_i} = \alpha_1 \langle x, y_1\rangle + \alpha_2 \langle x, y_2\rangle.$

#2 a) $\|x\| = \sqrt{\langle x, x\rangle} \geq 0$

$|x| = 0 \iff \sqrt{\langle x, x\rangle} = 0 \iff \langle x, x\rangle = 0 \iff x = 0$

b) $|\alpha x| = \sqrt{\langle \alpha x, \alpha x\rangle} = \sqrt{\alpha^2 \langle x, x\rangle} = |\alpha| \cdot \sqrt{\langle x, x\rangle} = |\alpha| \cdot |x|$.

#3 a) $|x| = |x + 0| = |x + (y - y)| = |(x - y) + y| \leq |x - y| + |y| \Rightarrow |x| - |y| \leq |x - y|$

$|y| = |y + 0| = |y + (x - x)| = |(y - x) + x| \leq |y - x| + |x| = |x - y| + |x| \Rightarrow -(|x| - |y|) \leq |x - y|$

b) $|x + y|^2 + |x - y|^2 = \langle x + y, x + y\rangle + \langle x - y, x - y\rangle = [\langle x, x\rangle + 2\langle x, y\rangle + \langle y, y\rangle]$

$+ [\langle x, x\rangle - 2\langle x, y\rangle + \langle y, y\rangle] = 2\langle x, x\rangle + 2\langle y, y\rangle = 2|x|^2 + 2|y|^2$

#4 Si $m = 1$, c'est l'inégalité du triangle. Supp. donc vrai pour m.
Alors, $|x_1 + \cdots + x_m + x_{m+1}| = |(x_1 + \cdots + x_m) + x_{m+1}| \leq |x_1 + \cdots + x_m| + |x_{m+1}|$
$\leq (|x_1| + \cdots + |x_m|) + |x_{m+1}|$

#5

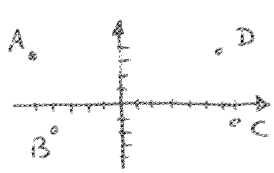
	(a)	(b)	(c)	(d)
i)	$(-1, -2)$	$(3, 1)$	$(-2, 2)$	$(a, -b)$
ii)	$(1, 2)$	$(-3, -1)$	$(2, -2)$	$(-a, b)$
iii)	$(1, -2)$	$(-3, 1)$	$(2, 2)$	$(-a, -b)$

#6 $|\overrightarrow{AB}| = |(8, 6) - (2, 3)| = |(6, 3)| = \sqrt{45} = 3\sqrt{5}$

$|\overrightarrow{AC}| = |(0, 7) - (2, 3)| = |(-2, 4)| = \sqrt{20} = 2\sqrt{5}$

$|\overrightarrow{BC}| = |(0, 7) - (8, 6)| = |(-8, 1)| = \sqrt{65}$

#7



$AB = (1, -5) \quad DC = (1, -5) \quad BC = (11, 1) \quad AD = (11, 1)$

$\overrightarrow{AB} \parallel \overrightarrow{DC}$ pcq. $\cos\alpha = \dfrac{\langle AB, DC\rangle}{|AB| \cdot |DC|} = 1$

$\overrightarrow{BC} \parallel \overrightarrow{AD}$ pcq. $\cos\alpha = 1$.

#8 $|\overrightarrow{AB}| = |(-3, -4)| = 5 \qquad |\overrightarrow{AC}| = |(4, 3)| = 5 \qquad |\overrightarrow{BC}| = |(7, 7)| = \sqrt{98} = 7\sqrt{2}$

C'est donc un triangle isocèle.

#9. $|\vec{AB}| = |(1,-6)| = \sqrt{37}$ $\quad |\vec{AC}| = |(-3,-4)| = 5$ $\quad |\vec{BC}| = |(-4,2)| = \sqrt{20}$.
Le périmètre est $\sqrt{37} + 5 + \sqrt{20}$.

#10. Posons $A(0,0)$, $B(2,2)$, $C(1,3)$ et $D(-1,1)$. On a $|\vec{AB}| = |(2,2)| = 2\sqrt{2}$, $|\vec{BC}| = |(-1,1)| = \sqrt{2}$, $|\vec{DC}| = |(2,2)| = 2\sqrt{2}$ et $|\vec{AD}| = |(-1,1)| = \sqrt{2}$.
De plus, $\langle \vec{AB}, \vec{BC} \rangle = \langle (2,2),(-1,1) \rangle = 0 \Rightarrow \vec{AB} \perp \vec{BC}$
C'est donc un rectangle.

#11. Posons $A(2,3)$, $B(5,7)$, $C(6,6)$, $D(-1,7)$ et $E(-2,0)$. On a
$|\vec{AB}| = |(3,4)| = 5$, $|\vec{AC}| = |(4,3)| = 5$, $|\vec{AD}| = |(-3,4)| = 5$ et $|\vec{AE}| = |(-4,-3)| = 5$.

#12 a) Posons $A(1,2)$, $B(9,6)$ et $C(-5,-1)$. On a $\dfrac{\langle \vec{AB}, \vec{AC} \rangle}{|\vec{AB}| \cdot |\vec{AC}|} =$
$\dfrac{\langle (8,4),(-6,-3) \rangle}{|(8,4)| \cdot |(-6,-3)|} = \dfrac{-60}{\sqrt{80}\sqrt{45}} = \dfrac{-60}{4\sqrt{5} \cdot 3\sqrt{5}} = -1$.
Donc, $\cos\alpha = -1$ où α est l'angle entre $\vec{AB}$ et $\vec{AC}$, d'où $\alpha = \pi$.

b) $\dfrac{\langle \vec{AB}, \vec{AC} \rangle}{|\vec{AB}| \cdot |\vec{AC}|} = \dfrac{\langle (3,5),(9,15) \rangle}{|(3,5)| \cdot |(9,15)|} = \dfrac{102}{\sqrt{34}\sqrt{306}} = 1 \Rightarrow \alpha = 0$

c) Posons $a=0$, $b=1$ et $c=2$. Alors, $A=(a, b-c) = (0,-1)$,
$B=(b, c-a) = (1,2)$ et $C=(c, a-b) = (2,-1)$. Ainsi,
$\dfrac{\langle \vec{AB}, \vec{AC} \rangle}{|\vec{AB}| \, |\vec{AC}|} = \dfrac{\langle (1,3),(2,0) \rangle}{|(1,3)| \cdot |(2,0)|} = \dfrac{2}{\sqrt{10} \cdot 2} = \dfrac{1}{\sqrt{10}}$. Les pts ne sont pas colinéaires.

#13 $\tan(\alpha_x - \alpha_y) = \dfrac{m_x - m_y}{1 + m_x m_y} \Rightarrow \tan\dfrac{\pi}{4} = \dfrac{3/2 - m_y}{1 + \frac{3}{2} m_y} \Rightarrow \left(1 + \frac{3}{2} m_y\right) = \frac{3}{2} - m_y$

$\Rightarrow \frac{5}{2} m_y = \frac{1}{2} \Rightarrow m_y = 1/5$.

#14 a) Posons $A(1,-3)$, $B(1,8)$ et $C(6,7)$. On a $\vec{AB} = (0,11)$, $\vec{AC} = (5,10)$ et $\vec{BC} = (5,-1)$.
$\vec{BA} = (0,-11)$, $\vec{CA} = (-5,-10)$ et $\vec{CB} = (-5,1)$

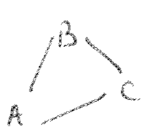
Attention à prendre la bonne orientation des vecteurs

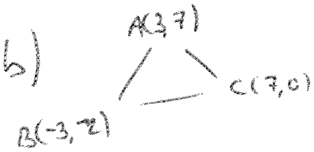
$\dfrac{\langle \vec{AB}, \vec{AC} \rangle}{|\vec{AB}| |\vec{AC}|} = \dfrac{110}{11 \cdot \sqrt{125}} = \dfrac{10}{5\sqrt{5}} = \dfrac{2}{\sqrt{5}} \Rightarrow \alpha = \arccos\dfrac{2}{\sqrt{5}} \approx 26{,}6°$

$\dfrac{\langle \vec{BA}, \vec{BC} \rangle}{|\vec{BA}| |\vec{BC}|} = \dfrac{11}{11 \cdot \sqrt{26}} = \dfrac{1}{\sqrt{26}} \Rightarrow \beta \approx 78{,}7°$

$\dfrac{\langle \vec{CA}, \vec{CB} \rangle}{|\vec{CA}| \cdot |\vec{CB}|} = \dfrac{15}{\sqrt{125} \cdot \sqrt{26}} = \dfrac{3}{\sqrt{130}} \Rightarrow \gamma \approx 74{,}7°$

$\alpha + \beta + \gamma \approx 180°$.



b) 

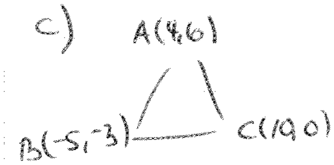
$\vec{AB} = (-6,-9)$ $\quad$ $\vec{BA} = (6,9)$ $\quad$ $\vec{CA} = (-4,7)$

$\vec{AC} = (4,-7)$ $\quad$ $\vec{BC} = (10,2)$ $\quad$ $\vec{CB} = (-10,-2)$

$$\cos\angle(\vec{AB},\vec{AC}) = \frac{39}{\sqrt{117}\cdot\sqrt{65}} = \frac{3\sqrt{13}}{3\cdot\sqrt{5}\sqrt{13}} = \frac{1}{\sqrt{5}} \implies \alpha \approx 63{,}4^\circ$$

$$\cos\angle(\vec{BA},\vec{BC}) = \frac{78}{\sqrt{117}\sqrt{104}} = \frac{2\cdot 3\cdot 13}{\sqrt{9\cdot 13}\sqrt{8\cdot 13}} = \frac{2}{\sqrt{8}} = \frac{1}{\sqrt{2}} \implies \beta = 45^\circ$$

$$\cos\angle(\vec{CA},\vec{CB}) = \frac{26}{\sqrt{65}\sqrt{104}} = \frac{2\cdot 13}{\sqrt{5\cdot 13}\cdot\sqrt{8\cdot 13}} = \frac{1}{\sqrt{10}} \implies \gamma \approx 71{,}6^\circ$$

c) 

$\vec{AB} = (-9,-9)$ $\quad$ $\vec{BA} = (9,9)$ $\quad$ $\vec{CA} = (-6,6)$

$\vec{AC} = (6,-6)$ $\quad$ $\vec{BC} = (15,3)$ $\quad$ $\vec{CB} = (-15,-3)$

$$\cos\angle(\vec{AB},\vec{AC}) = 0 \implies \alpha = 90^\circ$$

$$\cos\angle(\vec{BA},\vec{BC}) = \frac{162}{\sqrt{162}\sqrt{234}} = \sqrt{\frac{162}{234}} = \sqrt{\frac{9\cdot 9\cdot 2}{9\cdot 13\cdot 2}} = \frac{3}{\sqrt{13}} \implies \beta \approx 33{,}7^\circ$$

$$\cos\angle(\vec{CA},\vec{CB}) = \frac{72}{\sqrt{72}\sqrt{234}} = \sqrt{\frac{6\cdot 6\cdot 2}{9\cdot 13\cdot 2}} = \frac{2}{\sqrt{13}} \implies \gamma \approx 56{,}3^\circ.$$

#15 $\tan(\alpha_x - \alpha_y) = \dfrac{m_x - m_y}{1 + m_x m_y} = \dfrac{m - 1/m}{1 + m\cdot 1/m} = \dfrac{m^2-1}{2m} \implies \alpha_x - \alpha_y = \arctan\dfrac{m^2-1}{2m}.$

#16 $13 = |x| = \sqrt{4^2 + (-12)^2 + \alpha^2} = \sqrt{160 + \alpha^2} \implies 160 + \alpha^2 = 169 \implies \alpha = \pm 3$

#17 $P_{e_1}(x) = \dfrac{\langle x, e_1\rangle}{\langle e_1, e_1\rangle} = \dfrac{|x|\cdot|e_1|\cdot\cos\alpha}{|e_1|^2} = |x|\cdot\cos\alpha = 2\cdot\cos\dfrac{\pi}{4} = \sqrt{2}$

$P_{e_2}(x) = |x|\cdot\cos\beta = 2\cdot\cos\pi/3 = 1$

$P_{e_3}(x) = |x|\cdot\cos\gamma = 2\cdot\cos\dfrac{2\pi}{3} = -1.$

#18 $\cos\alpha = \dfrac{x_1}{|x|} = \dfrac{12}{25}$ $\qquad$ $\cos\beta = \dfrac{-15}{25}$ $\qquad$ $\cos\gamma = \dfrac{-16}{25}.$

#19 a) Non, pcq $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = \frac{1}{2} + \frac{1}{2} + \frac{1}{4} \neq 1.$

b) Oui, $\cos\alpha = 0 \implies x_1 = 0$, $\cos\beta = \cos\frac{5\pi}{6} = \frac{-\sqrt{3}}{2} = x_2/|x|$ et $\cos\gamma = \cos\frac{\pi}{3}$ $= \frac{1}{2} = \frac{x_3}{|x|} \implies |x| = \frac{-2}{\sqrt{3}}x_2$ et $|x| = 2x_3 \implies \frac{-2}{\sqrt{3}}x_2 = 2x_3 \implies x_2 = -\sqrt{3}x_3.$

Il suffit alors de prendre $x = (0, -\sqrt{3}, 1).$

#20 $1=\cos^2\alpha+\cos^2\beta+\cos^2\gamma=\left(\frac{-1}{2}\right)^2+\cos^2\beta+\left(\frac{\sqrt{2}}{2}\right)^2\Rightarrow\cos^2\beta=\frac{1}{4}\Rightarrow\beta=\frac{\pi}{3}$

#21 $|x+y|=\sqrt{\langle x+y,x+y\rangle}=\sqrt{\langle x,x\rangle+2\langle x,y\rangle+\langle y,y\rangle}=\sqrt{|x|^2+2|x|\cdot|y|\cdot\cos\alpha+|y|^2}$
$=\sqrt{12^2+2\cdot12\cdot5\cdot\cos 90^\circ+5^2}=13$ et $|x-y|=\sqrt{|x|^2-2|x|\cdot|y|\cdot\cos\alpha+|y|^2}=13$.

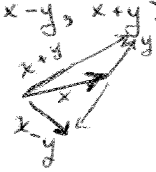
#22 a) $24=|x+y|=\sqrt{|x|^2+2|x|\cdot|y|\cdot\cos\alpha+|y|^2}=\sqrt{13^2+2\cdot13\cdot19\cdot\cos\alpha+19^2}\Rightarrow$
$576=169+494\cos\alpha+361\Rightarrow\cos\alpha=\frac{46}{494}$. Alors, $|x-y|=$
$\sqrt{|x|^2-2\cdot|x|\cdot|y|\cdot\cos\alpha+|y|^2}=\sqrt{13^2-2\cdot13\cdot19\cdot\frac{46}{494}+19^2}=\sqrt{484}=22$

b) $30=|x-y|=\sqrt{|x|^2-2|x||y|\cos\alpha+|y|^2}=\sqrt{1^2-2\cdot1\cdot23\cdot\cos\alpha+23^2}=\sqrt{530-46\cos\alpha}$
$\Rightarrow 900=530-46\cos\alpha\Rightarrow\cos\alpha=\frac{-370}{46}<-1$. De tels vecteurs n'existent pas.

#23 $\langle x,\alpha_1y_1+\dots+\alpha_my_m\rangle=\alpha_1\langle x,y_1\rangle+\dots+\alpha_m\langle x,y_m\rangle=\alpha_1\cdot0+\dots+\alpha_m\cdot0=0$

#24 $\langle x,y\rangle=\langle(1,2,3,4),(0,3,-2,1)\rangle=4$, $|x|=\sqrt{30}$, $|y|=\sqrt{14}$
$\cos\alpha=\frac{4}{\sqrt{30}\sqrt{14}}=\frac{2}{\sqrt{105}}\Rightarrow\alpha\approx78{,}7^\circ$

#25 $\langle x-y,x+y\rangle=\langle x,x\rangle+\langle x,y\rangle-\langle y,x\rangle-\langle y,y\rangle=\langle x,x\rangle-\langle y,y\rangle=|x|^2-|y|^2=0$.



#26 Posons $x=(x_1,x_2,x_3,x_4)$. On a $x\perp(3,2,2,4)\Rightarrow3x_1+2x_2+2x_3+4x_4=0$,
$x\perp(1,0,0,2)\Rightarrow x_1+2x_4=0$ et $x\perp(1,-1,-1,1)\Rightarrow x_1-x_2-x_3+x_4=0$. Ainsi,
$\begin{pmatrix}3&2&2&4\\1&0&0&2\\1&-1&-1&1\end{pmatrix}\sim\begin{pmatrix}1&0&0&2\\0&-1&-1&-1\\0&2&2&-2\end{pmatrix}\sim\begin{pmatrix}1&0&0&2\\0&1&1&1\\0&0&0&-4\end{pmatrix}\sim\begin{pmatrix}1&0&0&0\\0&1&1&0\\0&0&0&1\end{pmatrix}$. On en déduit que $x_1=0=x_4$ et $x_2=-x_3$.
Comme $|x|=1$, on trouve $x=\left(0,\frac{\sqrt{2}}{2},\frac{-\sqrt{2}}{2},0\right)$.

#27 $|x+y|=\sqrt{|x|^2+2|x|\cdot|y|\cdot\cos\alpha+|y|^2}=\sqrt{5^2+2\cdot5\cdot8\cdot\cos\frac{\pi}{3}+8^2}=\sqrt{129}$
$|x-y|=\sqrt{|x|^2-2|x|\cdot|y|\cos\alpha+|y|^2}=\sqrt{5^2-2\cdot5\cdot8\cdot\cos\frac{\pi}{3}+8^2}=7$.

#28 $\langle x,y\rangle+\langle y,z\rangle+\langle z,x\rangle=\langle x,(x+y+z)-(x+z)\rangle+\langle y,(x+y+z)-(x+y)\rangle+\langle z,(x+y+z)-(y+z)\rangle$
$=\langle x,x+y+z\rangle-\langle x,x+z\rangle+\langle y,x+y+z\rangle-\langle y,x+y\rangle+\langle z,x+y+z\rangle-\langle z,y+z\rangle$
$=-\langle x,x\rangle-\langle x,z\rangle-\langle y,x\rangle-\langle y,y\rangle-\langle z,y\rangle-\langle z,z\rangle=-|x|^2-|y|^2-|z|^2-\langle z,x\rangle-\langle x,y\rangle-\langle y,z\rangle$
$\Rightarrow2\langle x,y\rangle+2\langle y,z\rangle+2\langle z,x\rangle=-|x|^2-|y|^2-|z|^2=-3$ d'où le résultat.

#29 $0 = \langle x+\alpha y, x-\alpha y\rangle = \langle x,x\rangle - \alpha\langle x,y\rangle + \alpha\langle y,x\rangle - \alpha^2\langle y,y\rangle = |x|^2 - \alpha^2|y|^2$
$= 3^2 - \alpha^2 \cdot 5^2 = 9 - 25\alpha^2 \Rightarrow \alpha^2 = 9/25 \Rightarrow \alpha = \pm 3/5$.

#30 $|x+y| = \sqrt{|x|^2 + 2|x|\cdot|y|\cdot\cos\alpha + |y|^2} = \sqrt{3^2 + 2\cdot 3\cdot 5\cdot\cos\frac{2\pi}{3} + 5^2} = \sqrt{19}$
$|x-y| = \sqrt{3^2 - 2\cdot 3\cdot 5\cdot\cos\frac{2\pi}{3} + 5^2} = 7$

#31 $|x+y| = |x-y| \Leftrightarrow |x|^2 + |y|^2 + 2|x||y|\cos\alpha = |x|^2 + |y|^2 - 2|x|\,|y|\cos\alpha$
$\Leftrightarrow \cos\alpha = 0 \Leftrightarrow \alpha = \frac{\pi}{2}$

#32 $\cos\alpha = \dfrac{\langle x,y\rangle}{|x|\cdot|y|} = \dfrac{\langle (2,-1,3),(-6,3,-9)\rangle}{|(2,-1,3)|\cdot|(-6,3,-9)|} = \dfrac{-42}{\sqrt{14}\cdot\sqrt{126}} = -1 \Rightarrow \alpha = \pi$.

#33 Si deux vect. sont colin., leurs cos directeurs sont égaux, à signe près.
Ainsi, si on pose $x = (-2, 3, \beta)$ et $y = (\alpha, -6, 2)$, on a que
$\frac{-2}{|x|} = \pm\frac{\alpha}{|y|}$, $\frac{3}{|x|} = \pm\frac{-6}{|y|}$ et $\frac{\beta}{|x|} = \pm\frac{2}{|y|}$. De la 2e égalité, on tire $|y| = \pm -2|x|$.
et on choisit $|y| = 2|x|$.
Les deux autres égalités sont alors $\frac{-2}{|x|} \pm \frac{\alpha}{2|x|} \Rightarrow \alpha = \pm 4$ et $\frac{\beta}{|x|} = \frac{\pm 2}{2|x|} \Rightarrow \beta = \pm 1$.
On vérifie aisément que $\alpha = 4$ et $\beta = -1$ sont les valeurs cherchées.

#34 Du numéro 28, on tire $\langle x,y\rangle + \langle y,z\rangle + \langle z,x\rangle = -\frac{1}{2}\left[|x|^2 + |y|^2 + |z|^2\right] =$
$-\frac{1}{2}\left[3^2 + 1^2 + 4^2\right] = -\frac{1}{2}[26] = -13$.

#35 $x+y \perp x-y \Leftrightarrow \langle x+y, x-y\rangle = 0 \Leftrightarrow |x|^2 - |y|^2 = 0 \Leftrightarrow |x|^2 = |y|^2 \Leftrightarrow |x| = |y|$.

#36 $\cos\alpha = \dfrac{\langle x+y, x-y\rangle}{|x+y|\cdot|x-y|} = \dfrac{|x|^2 - |y|^2}{\sqrt{|x|^2 + |y|^2 + 2|x||y|\cos\alpha}\sqrt{|x|^2 + |y|^2 - 2|x||y|\cos\alpha}} = \dfrac{2}{\sqrt{3+1+2\sqrt{3}\cdot\frac{\sqrt{3}}{2}}\sqrt{3+1-2\sqrt{3}\frac{\sqrt{3}}{2}}}$
$= \dfrac{2}{\sqrt{7}\sqrt{1}} = \dfrac{2}{\sqrt{7}} \Rightarrow \alpha \approx 40,9°$

#37 Posons $x = (x_1, x_2, \ldots, x_n)$ et $u = (u_1, \ldots, u_n)$. Si $\alpha = 0$ et $u \neq 0$, ce sont les vect. orthogonaux à u, c-à-d. le disque centré à l'origine au vect. u. Si $u = 0$ et $\alpha = 0$, c'est l'espace au complet.
Supposons donc que $\alpha \neq 0$ et $u \neq 0$. Alors, $\langle u, x\rangle = \alpha$
$\Rightarrow \sum_{i=1}^{n} u_i x_i = \alpha \Rightarrow x_n = \dfrac{\alpha - \sum_{i=1}^{n-1} u_i x_i}{u_n}$

Supposons, SPDG, que $u_n \neq 0$

Ainsi, $\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_{n-1} \\ x_n \end{pmatrix} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_{n-1} \\ \frac{\alpha}{u_n} - \frac{u_1}{u_n}x_1 - \frac{u_2}{u_n}x_2 - \cdots - \frac{u_{n-1}}{u_n}x_{n-1} \end{pmatrix} = \underbrace{\begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ \frac{\alpha}{u_n} \end{pmatrix}}_{z_0} + x_1 \underbrace{\begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \\ \frac{-u_1}{u_n} \end{pmatrix}}_{z_1} + x_2 \underbrace{\begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \\ \frac{-u_2}{u_n} \end{pmatrix}}_{z_2} + \cdots + x_{n-1} \underbrace{\begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \\ \frac{-u_{n-1}}{u_n} \end{pmatrix}}_{z_{n-1}}$

Le lieu cherché est donc l'hyperplan engendré par les vecteurs $z_1, \ldots, z_{n-1}$ qui aura été translaté de z_0.

Par ex., dans $\mathbb{R}^3$, si $\alpha = 2$ et $u = (12, 8, 4)$, c'est le plan engendré par les vect. $z_1 = \begin{pmatrix} 1 \\ 0 \\ -3 \end{pmatrix}$ et $z_2 = \begin{pmatrix} 0 \\ 1 \\ -2 \end{pmatrix}$, translaté de $z_0 = \begin{pmatrix} 0 \\ 0 \\ 1/2 \end{pmatrix}$.

#38 $\vec{AC} = (5,3,-1)$, $\vec{BD} = (-6,-9,3) \Rightarrow \langle \vec{AC}, \vec{BD} \rangle = 30 - 27 - 3 = 0$

#39 $(\alpha, -3, 2) \perp (1, 2, -\alpha) \Leftrightarrow \alpha - 6 - 2\alpha = 0 \Rightarrow -6 - \alpha = 0 \Rightarrow \alpha = -6$.

#40 $\cos\alpha = \frac{\langle x, y \rangle}{|x| \cdot |y|} = \frac{\langle (2,-4,4), (-3,2,6) \rangle}{|(2,-4,4)| \cdot |(-3,2,6)|} = \frac{10}{6 \cdot 7} = \frac{10}{42} = \frac{5}{21}$

#41 $\vec{BA} = (3, 0, -2)$ et $\vec{BC} = (7, 0, 1) \Rightarrow \cos\alpha = \frac{\langle (3,0,-2), (7,0,1) \rangle}{|(3,0,-2)| \cdot |(7,0,1)|} = \frac{19}{\sqrt{13}\sqrt{50}} = \frac{19}{\sqrt{650}} \Rightarrow \alpha \approx 41{,}8^\circ$

#42 $\vec{AB} = (2,-1,2)$, $\vec{AC} = (-2,-4,2) \Rightarrow \cos\alpha = \frac{4}{3 \cdot \sqrt{24}} = \frac{2}{3\sqrt{6}} \Rightarrow \alpha \approx 74{,}2^\circ$ d'où $285{,}8^\circ$

#43 $|\vec{AB}| = |(2,-3,6)| = \sqrt{49} = 7$, $|\vec{AC}| = |(6, 2, -3)| = 7$.

#44 $\left.\begin{array}{r} x \perp (2,3,-1) \Rightarrow 2x_1 + 3x_2 - x_3 = 0 \\ x \perp (-1,2,3) \Rightarrow -x_1 + 2x_2 + 3x_3 = 0 \\ \langle x, (2,-1,1) \rangle = -6 \Rightarrow 2x_1 - x_2 + x_3 = -6 \end{array}\right\} \Rightarrow \begin{pmatrix} 2 & 3 & -1 \\ -1 & 2 & 3 \\ 2 & -1 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ -6 \end{pmatrix} \Rightarrow \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \frac{1}{17} \begin{pmatrix} -33 \\ 15 \\ -21 \end{pmatrix}$

#45 $\left.\begin{array}{l} x \perp e_1 \Rightarrow x_1 = 0 \\ \langle x, (3,-1,5) \rangle = 9 \Rightarrow 3x_1 - x_2 + 5x_3 = 9 \\ \langle x, (1,2,-3) \rangle = 4 \Rightarrow x_1 + 2x_2 - 3x_3 = 4 \end{array}\right\} \Rightarrow \begin{pmatrix} 1 & 0 & 0 \\ 3 & -1 & 5 \\ 1 & 2 & -3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 9 \\ 4 \end{pmatrix} \Rightarrow \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 47/7 \\ 22/7 \end{pmatrix}$

#46 $\langle x, (-2,1,3) \rangle = -5 \Rightarrow -2x_1 + x_2 + 3x_3 = -5$

$\langle x, (1,-3,2) \rangle = -11 \Rightarrow x_1 - 3x_2 + 2x_3 = -11$

$\langle x, (3,2,-4) \rangle = 20 \Rightarrow 3x_1 + 2x_2 - 4x_3 = 20$

$$\Rightarrow \begin{pmatrix} -2 & 1 & 3 \\ 1 & -3 & 2 \\ 3 & 2 & -4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -5 \\ -11 \\ 20 \end{pmatrix} \Rightarrow \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \frac{1}{27} \begin{pmatrix} 70 \\ 101 \\ -32 \end{pmatrix}$$

#48 $1 = \tan \pi/4 = \tan(\alpha_x - \alpha_y) = \dfrac{m_x - m_y}{1 + m_x m_y} \iff 1 + m_x m_y = m_x - m_y$

$\iff m_x - m_x m_y - m_y = 1.$